

SMTP, POP & IMAP

Lesson 38 — Mail Protocols and Threats

JKSSB Computer Awareness

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Why Email Needs Protocols

- ▶ Sending, receiving, and reading mail involve different mail servers and different programs.
- ▶ **Protocols** are fixed rules so any mail program can talk to any mail server.
- ▶ One protocol **sends** mail; two other protocols **receive** it in different ways.
- ▶ Secure transmission and threat awareness sit on top of these protocols.

The one idea

SMTP sends mail out; POP3 and IMAP bring it in — in two different ways.

SMTP: The Sending Protocol

- ▶ **SMTP** stands for Simple Mail Transfer Protocol.
- ▶ It is the protocol used to **send** mail from the sender's program to the mail server, and between mail servers.
- ▶ The sender's program hands the message to its SMTP server; that server forwards it toward the recipient's mail server.
- ▶ SMTP does **not** deliver mail to the recipient's inbox for reading — receiving is the job of POP3 or IMAP.

Direction to remember

SMTP is for **sending**: out of the sender, toward the recipient's server.

POP3: Download to One Machine

- ▶ **POP3** stands for Post Office Protocol version 3.
- ▶ It is a **receiving** protocol: it pulls mail from the mail server down to one local machine.
- ▶ After downloading, the mail is typically **removed from the server**, so it lives on that one computer.
- ▶ It suits a single computer where mail is read offline in one place.

Keep the pair straight

POP3 **downloads** mail to the local machine; it does not keep it synced on the server.

IMAP: Sync Across Devices

- ▶ **IMAP** stands for Internet Message Access Protocol.
- ▶ It is also a **receiving** protocol, but it works differently from POP3.
- ▶ Mail **stays on the server**; the program shows a synced view of folders, read marks, and deletions.
- ▶ Reading, deleting, or moving a message on the phone shows the same change on the laptop, because every device mirrors the server.

Office desktop example

Checking office mail on the phone and later on the desktop shows the same inbox — that is IMAP keeping every device in step with the server.

POP3 vs IMAP

Point	POP3	IMAP
Job	Receives mail	Receives mail
Method	Downloads to one machine	Syncs a view of the server
Server copy	Usually removed	Kept on the server
Many devices	Poor fit	Designed for it

Trap alert

POP3 **downloads** to local storage; IMAP **syncs** with the server. Do not swap them.
(TRAP-17)

Mail-Server Flow, End to End

- ▶ The sender writes a message and the mail program sends it by **SMTP** to the sender's mail server.
- ▶ The sender's server forwards it by **SMTP** to the recipient's mail server, where it waits in the mailbox.
- ▶ The recipient's program collects it with **POP3** (download) or **IMAP** (sync).
- ▶ Sending is always SMTP; the receiving side is POP3 or IMAP.

The flow in one line

Sender → SMTP → recipient server → POP3 or IMAP → reader.

To, Cc, and BCC

- ▶ **To** names the main recipients; **Cc** (carbon copy) names others kept in the loop.
- ▶ **BCC** stands for blind carbon copy: its recipients are **hidden from each other**.
- ▶ Everyone sees the To and Cc lists; nobody on the To or Cc list sees who is on BCC.
- ▶ The **sender** always knows all recipients; BCC hides recipients from each other, not from the sender.

Trap alert

BCC hides recipients **from each other**. (TRAP-16)

Filters, Rules, and Signatures

- ▶ **Filters (rules)** sort incoming mail automatically — for example, moving newsletters to one folder or flagging important senders.
- ▶ A filter acts on the **headers and content** of the message, such as sender, subject, or words in the body.
- ▶ A mail **signature** is a fixed block (name, post, phone) added at the end of outgoing messages.
- ▶ A signature identifies the sender; it does **not** encrypt or protect the message.

Keep the pair straight

Filters organise incoming mail; a signature labels outgoing mail. Neither one encrypts anything.

Digital Signature vs Encryption

- ▶ A **digital signature** proves who sent the message and that it was not changed — authenticity and integrity.
- ▶ **Encryption** scrambles the message so only the intended recipient can read it — confidentiality.
- ▶ A signed message can still be **read** by anyone; a signed message is not automatically secret.
- ▶ An encrypted message can still be **forwarded**; encryption hides content, it does not prove the sender.

Trap alert

A digital signature does **not** encrypt the body. Signature = authenticity; encryption = secrecy. (TRAP-15)

Secure Email: SSL and TLS

- ▶ **SSL** stands for Secure Sockets Layer; **TLS** stands for Transport Layer Security.
- ▶ Together **SSL/TLS** provide the secure channel over which mail is transmitted.
- ▶ Without this protection, mail travels in a form others on the path could read.
- ▶ Exam answer: the secure email transmission protocol is **SSL/TLS**. (PYQ-36)

Exam recall

Secure transmission of email = SSL/TLS.

Spam

- ▶ **Spam** is bulk unwanted mail, usually advertising or junk sent to many addresses at once.
- ▶ It clutters the inbox and can carry fraud or malware, but the defining mark is **bulk and unwanted**.
- ▶ Spam filters move suspected junk to a separate folder using sender reputation and content rules.
- ▶ Deleting spam without opening links or attachments is the safe habit.

Office desktop example

Dozens of identical “prize winner” messages arriving overnight go to the junk folder — that is spam filtering at work.

Phishing

- ▶ **Phishing** is a fake message that pretends to be a trusted sender — a bank, an office, or a colleague — to steal passwords or money.
- ▶ It creates **urgency**: “verify now”, “account blocked”, “claim within 24 hours”.
- ▶ It asks the reader to click a link or open an attachment and hand over secrets.
- ▶ Check the real sender address and never enter passwords from a mail link.

Takeaway

Phishing = impersonation + urgency + a link or attachment that steals secrets.

Spoofing, Malicious Links, and Attachments

- ▶ **Spoofing** forges the sender address so a message looks as if it came from someone trusted.
- ▶ A **malicious link** leads to a fake login page or a malware download; hovering shows an address that does not match the claimed sender.
- ▶ A **malicious attachment** carries malware behind an innocent-looking file name.
- ▶ Rule: verify the sender by another channel before clicking or opening anything unexpected.

Keep the pair straight

Phishing is the **fraud attempt**; spoofing is the **forged sender** trick it often uses.

Trap Alert: Mail Facts to Keep Straight

- ▶ SMTP **sends**; POP3 and IMAP **receive**.
- ▶ POP3 **downloads** to one machine; IMAP **syncs** with the server. (TRAP-17)
- ▶ A digital signature proves the sender; it does **not** encrypt the message. (TRAP-15)
- ▶ BCC hides recipients **from each other**, not from the sender.
- ▶ Secure email transmission = **SSL/TLS**. (PYQ-36)

Lesson 38: Recall

- ▶ SMTP (Simple Mail Transfer Protocol) sends mail; POP3 (Post Office Protocol version 3) downloads it to one machine; IMAP (Internet Message Access Protocol) syncs it across devices.
- ▶ Mail flow: sender → SMTP → recipient server → POP3 or IMAP → reader.
- ▶ BCC hides recipients from each other; filters and rules sort incoming mail; a signature labels outgoing mail.
- ▶ Digital signature = authenticity and integrity; encryption = confidentiality; secure transmission = SSL/TLS.
- ▶ Spam is bulk unwanted mail; phishing impersonates trust to steal secrets; spoofing forges the sender; malicious links and attachments deliver the fraud or malware.